The Invisible Wall: A Study of the 107% Rule in Formula 1

Introduction

The 107% rule has typically been considered the fixed measurement standard in Formula 1 that determines whether a car qualifies or is excluded (FiA, 2024). However, due to the current climate of marginal gains, there exists a more complex space between the front of the grid and the limit of regulations shaped by more than simply engine power. This project utilised visual analytics to break down this “invisible wall”. By examining how historical changes in technology, impact of different racetracks and precise driver input converge to create the exact framework for regulatory failure.

1. Research Questions

Main Research question:

How effective is visual analytics in revealing the "invisible" competitive thresholds within Formula 1 through grid density mapping and granular driver performance data, against the 107% regulatory threshold?

Sub-question 1:

- How has the performance density of the grid changed with respect to the 107% threshold across different technology eras (i.e. V8 engines vs turbo-hybrid engines or Ground Effects)?

Sub-question 2:

- How much does this gap change as a result of circuit design (Street vs Permanent track), and what types of layouts pose the greatest risk regarding meeting the 107% "Invisible Wall" threshold?

Sub-question 3:

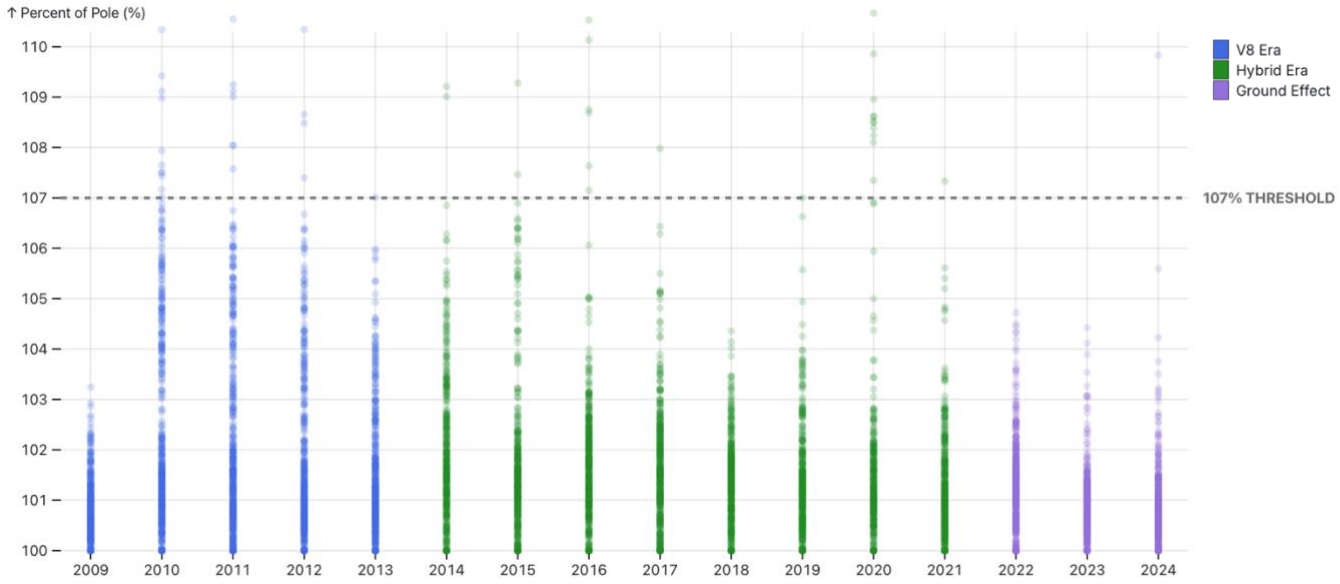
- How much do the variations in driver inputs account for the marginal amount of time that results in a driver’s failure in qualifying?

2. The Visualisations

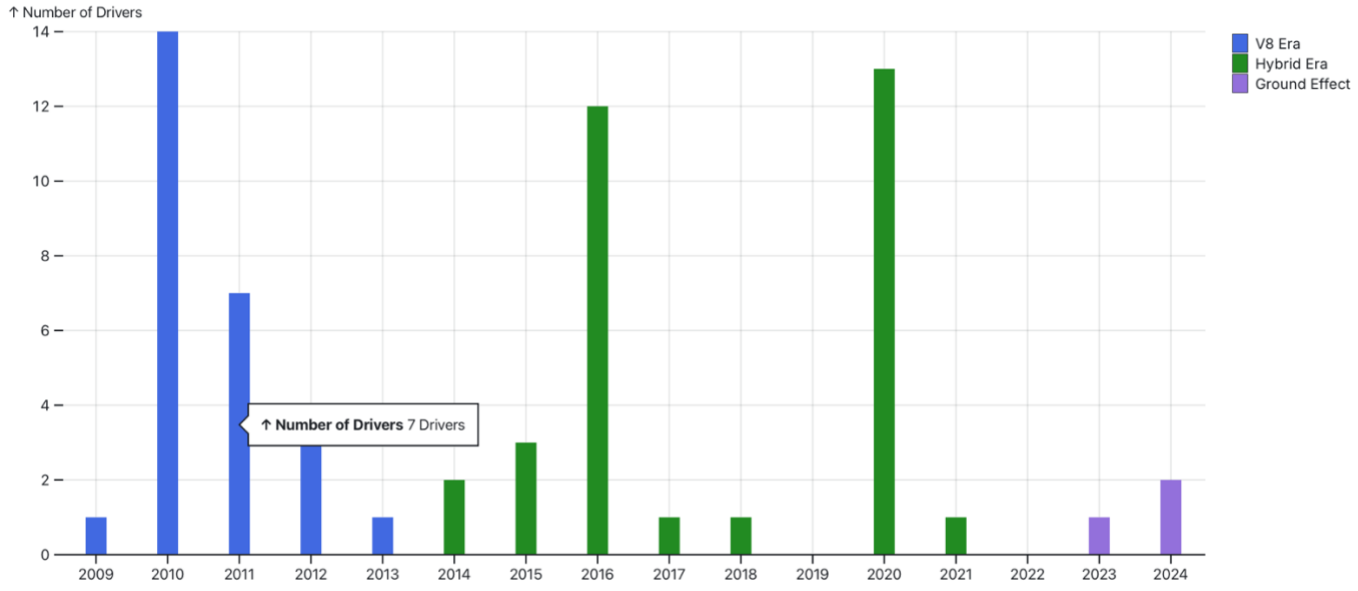
During the data preprocessing workflow, the Formula 1 World Championship (1950-2024) dataset from Kaggle was filtered to a 2009-2024 timeframe. The raw qualifying strings were then transformed into numbers representing seconds, which helped generate the 107% regulatory benchmark. Each track was grouped by either street or permanent track.

High-frequency sensor data from the OpenF1 API was also used to extract the telemetry data of Verstappen and Perez from the Qualifying 1 session of the 2024 Japanese Grand Prix. The collected data was cleaned by isolating the fastest-lap time windows and then using time-offset normalisation to line up the times from distinct timestamps and create proper comparisons between the speeds, throttle, brake, and gear measurements between the drivers.

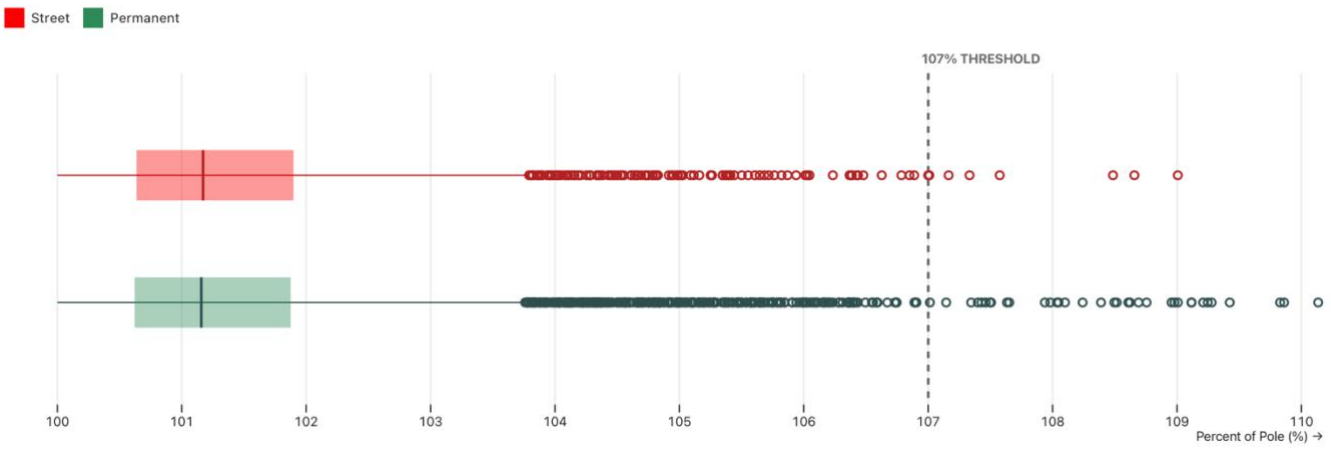
Driver Pace as % of the fastest time in Qualifying 1 per season



Count of Drivers Above 107% Threshold in Qualifying 1 per season

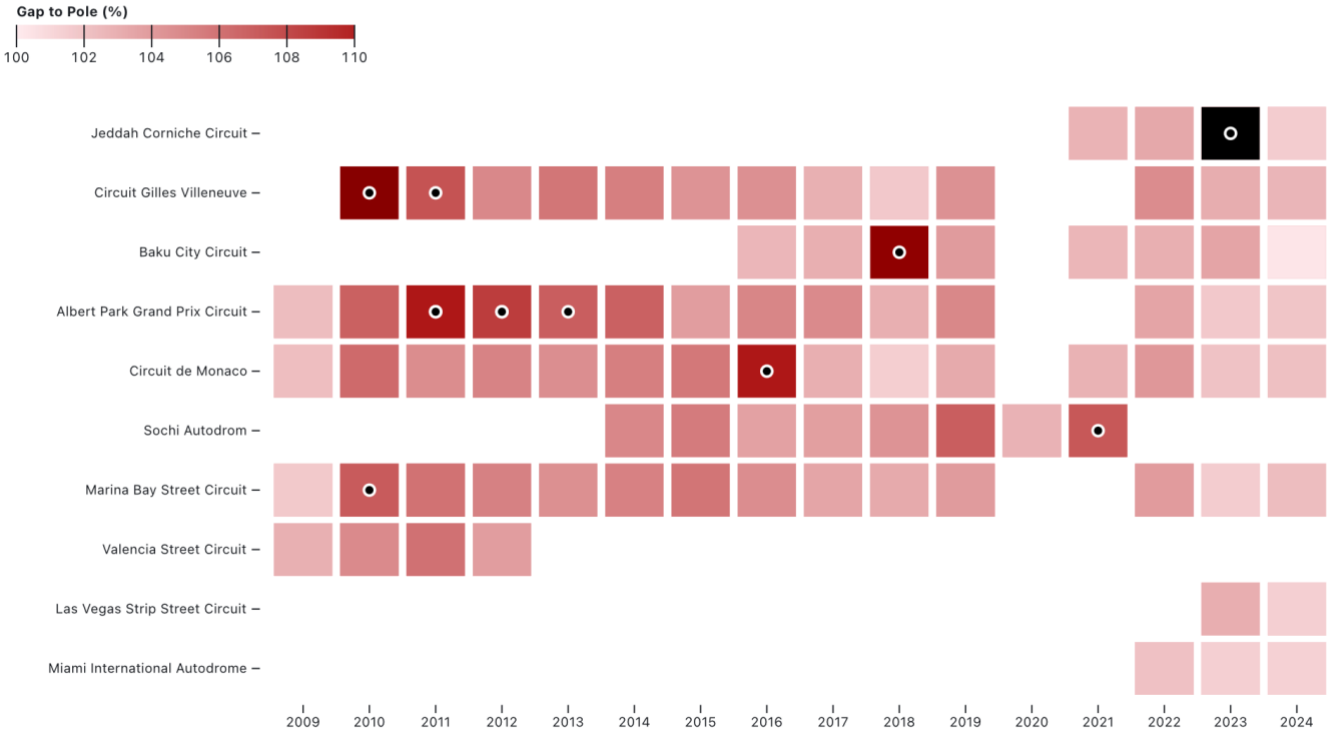


Impact of Track Type on the 107% Threshold



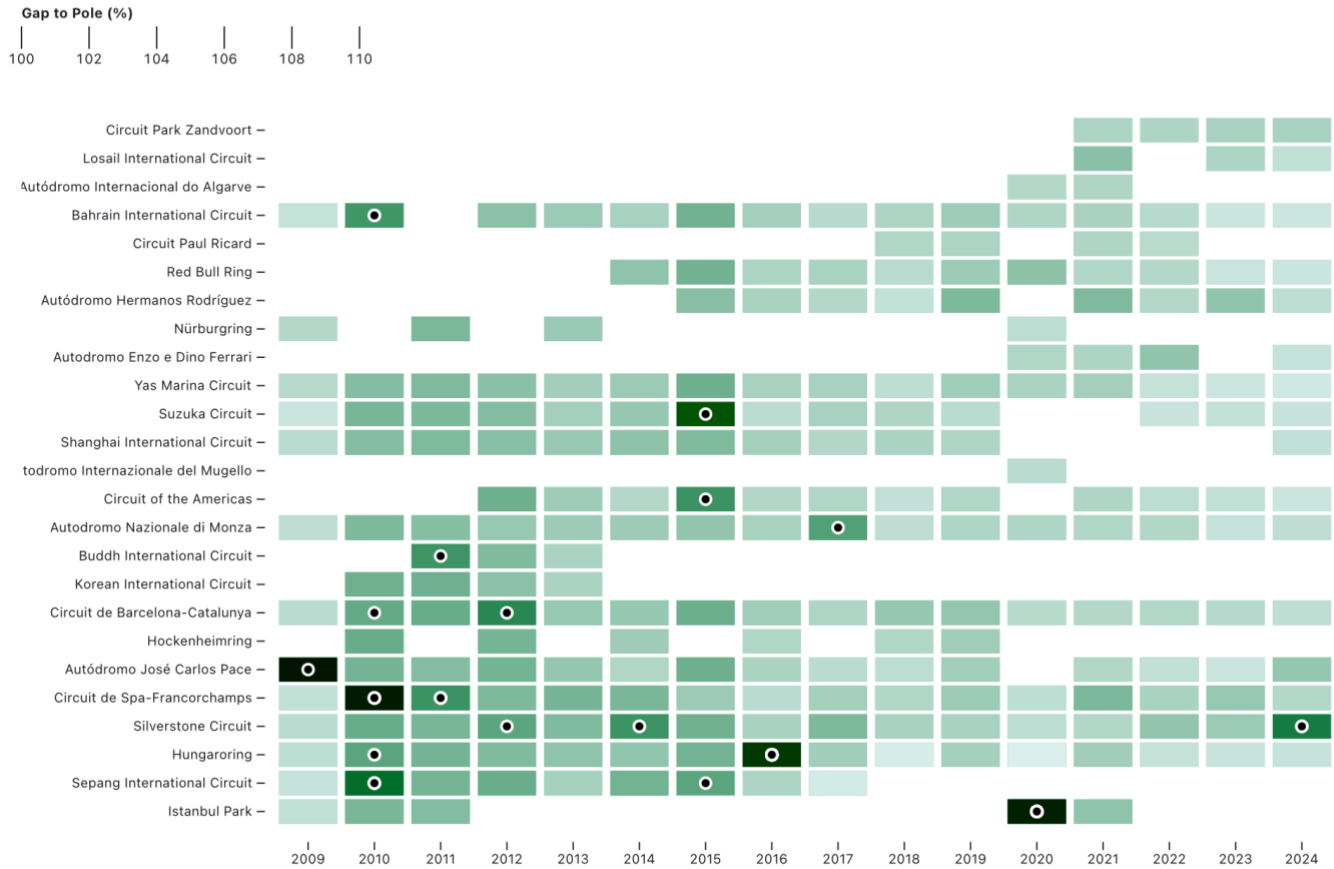
Street Circuits: 107% Regulatory Compliance During Qualifying 1

Darker hue with black dot = Regulatory Failure

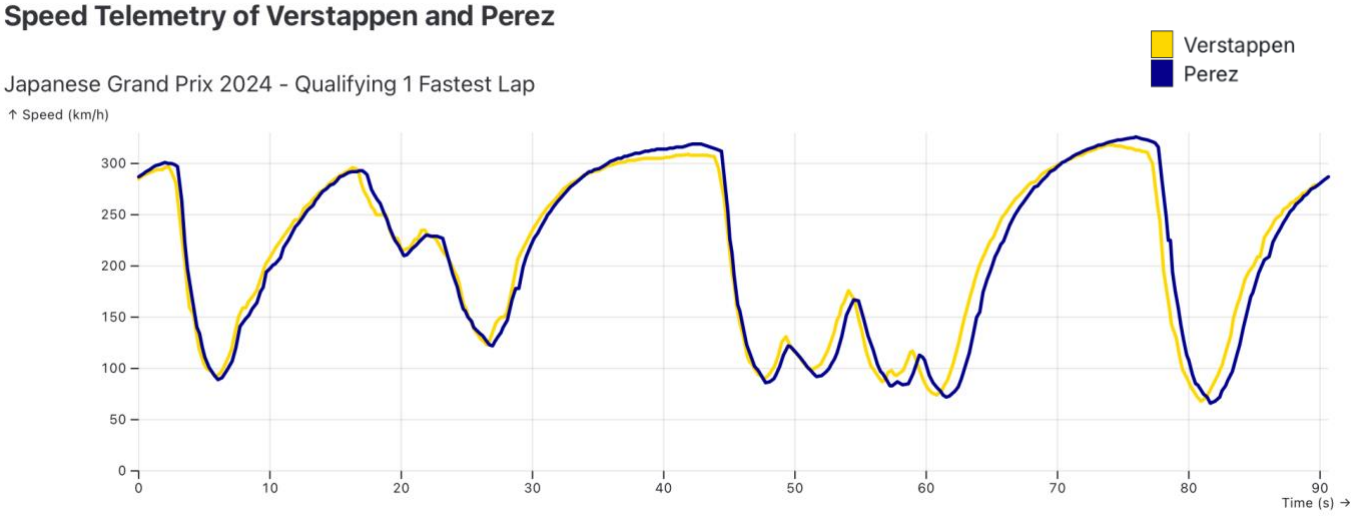


Permanent Circuits: 107% Regulatory Compliance During Qualifying 1

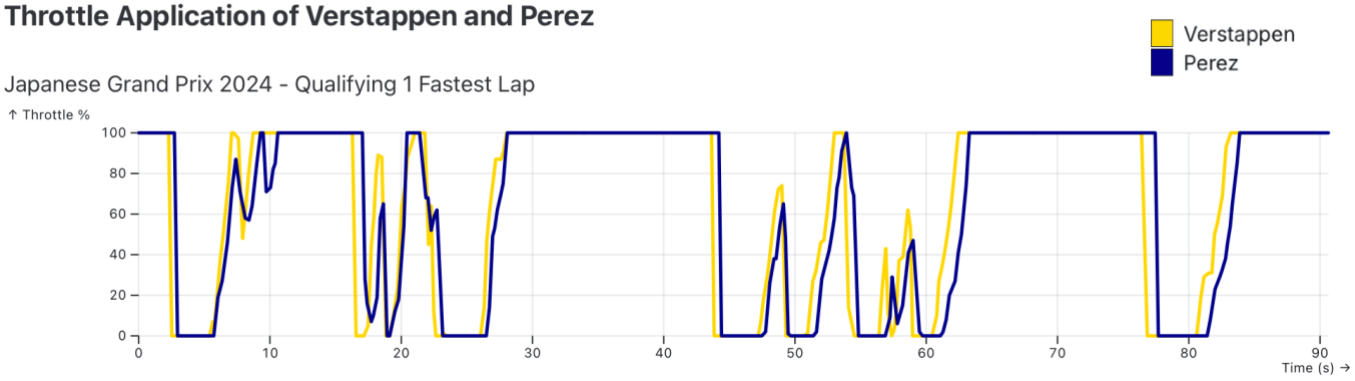
Darker hue with black dot = Regulatory Failure



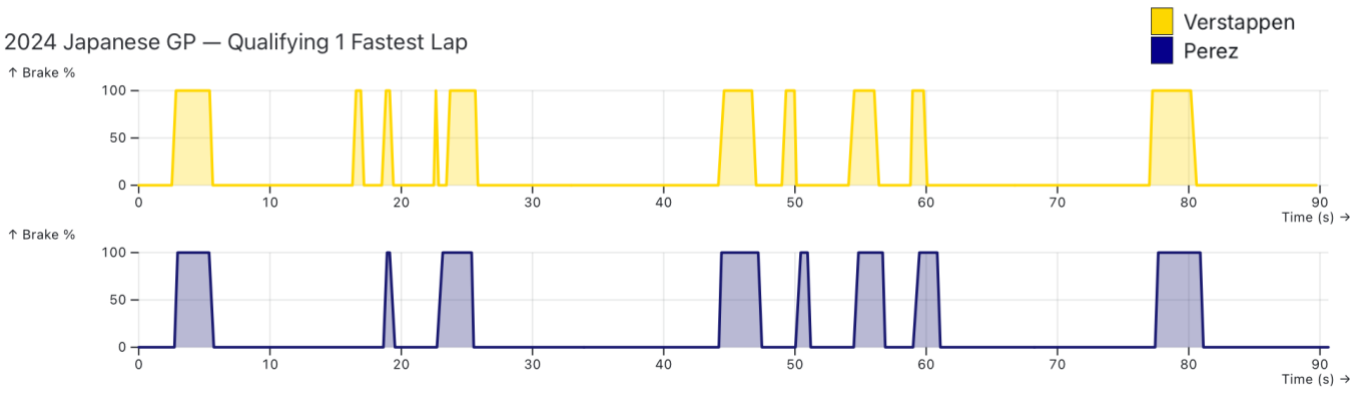
Speed Telemetry of Verstappen and Perez



Throttle Application of Verstappen and Perez



Braking Force of Verstappen and Perez



3. Insights

The longitudinal and comparative visualisations revealed the hidden architecture of the sport. The following three insights explain the impact of technical changes, circuit characteristics, and driver input on the 107% regulatory threshold:

The longitudinal scatter plot and bar graph reveal measurable effects of the transition between each engine type on the grid's level of gap from the pole. The V8 engine era and initial Hybrid era gaps were volatile, with the number of drivers above the threshold yearly ranging from 2 to 14. This likely reflects the technical experiments, such as blown diffusers and kinetic energy recovery systems, implemented by the teams. In the late Hybrid era, the average pole percentage lowered to around 102%, possibly due to the maturing power unit technologies. During the Ground Effect era, the pole percentage narrowed further towards 100%, potentially due to the restrictive regulations implemented. The evolution of engines over the years has transformed the 107% rule from being a valid competitive filter to being a nominal safety threshold, reducing the number of drivers above the regulatory threshold significantly.

Boxplots and heatmaps clearly show that while median gaps were fairly constant across track layouts, it was permanent circuits that drove most of the variability in gaps. High-speed circuits like Spa and Silverstone exhibited wider data dispersion in the box plot and concentrated clustered failures in the heatmap (4 failures over 4 years), revealing design and performance deficiencies that were invisible on street circuits. Conversely, failures at street circuits like Albert Park, with 3 consecutive years of regulation failures and Jeddah (gap to pole of 144.78%) could be due to a lack of their frequent use for motorsport events, affecting the track grip and evolution. While permanent circuits revealed the effect on the performance gap caused by sustained high-speed strain on the track, the regulatory failures in street circuits were primarily due to the inability of the teams to adapt the car according to the track's volatile conditions.

Telemetry data demonstrated a large gap between Perez and Verstappen during their fastest lap in Qualifying 1. Both drivers had the same 326 km/h top speed; however, Verstappen was faster than Perez in the majority of the track. Verstappen braked much earlier at 49s and in a more decisive manner, which allowed him to apply full throttle immediately when leaving the corner at 54.08s. Perez, on the other hand, was able to brake considerably later at 50.04s, followed by a hesitant application of the throttle at 54.48s. Thereby confirming that the invisible deficit is not due to a lack of power but a confidence deficiency. Such micro-hesitations in driver inputs can lead to an increase in qualifying times, resulting in >107% gap in qualifying.

Synthesis

The analysis ultimately shows the effectiveness of visual analytics in revealing Formula 1's "invisible" threshold. The use of grid density mapping and granular telemetry data demonstrates the value of moving beyond the binary classification of "pass or fail" and the idea that the 107% regulation is not static, instead representing a continually shifting border influenced by the technical maturity of the car, the geometry of the circuit and the drivers' confidence levels. With this multi-dimensional view, raw lap times become an effective diagnostic tool to identify areas of competitive imbalance.

4. Design Justification

The purpose of the design process was to expose how the "invisible" performance gaps lead to regulatory failures, while exploring design patterns to effectively visualise the motorsport database. Below are three design decisions made to address these objectives:

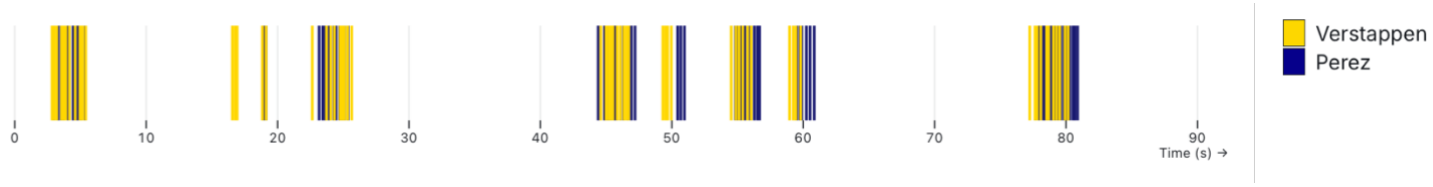
A concept-driven palette was implemented for the data categories based on how they're characterised and identified, which created intuitive access via Semantic Resonance (Lin et al., 2013). Forest green (Hybrid) was used to represent sustainability, medium purple for Ground Effect represented aerodynamic airflow, and royal blue for V8 represented high-technology engineering. Red represented the danger and high risk associated with street tracks, while the seagreen represented permanent circuit layouts in natural countryside. Gold was used for Verstappen, reflecting his position as a reigning world champion, and blue was used for Perez to represent the Red Bull Racing team for which he raced. The viewer's relationship to the data becomes richer through this method of affective mapping, where the viewer's prior mental models or emotional connections to the sport help enhance the viewer's visual experience. This colour mapping allows the user to reduce the cognitive load by using pre-attentive processing (Ware, 2004) to intuitively interpret the context of a performance gap prior to reviewing the legend.

Space juxtaposition was used to create a vertically uniform stacking arrangement (Gleicher et al., 2011) for the speed, throttle, and braking telemetry by aligning them on the x-axis. This structure relates to the small multiples idiom (Tufte, 1998) that allows viewers to conduct vertical scans to make connections between different data streams. For example, an occurrence of braking at approximately 49 seconds could be linked to the subsequent delay in throttle at 54 seconds. Since the overall stack is operating on the same temporal scale, it creates a more visible or apparent relationship between driver actions and their effect on vehicle behaviour, therefore preventing readers from getting overwhelmed by high-dimensional racing telemetric data.

A sequential colour gradient was implemented on the heatmaps, in which the gradient range was expanded from 100-107% to a functional value of 100-110%. This ensured that extreme regulatory outliers remain visible without being clipped out of the data, and the darker-hued boxes were more distinct for easier comparisons. Moreover, the black circles with white borders depicting the regulatory failures were implemented to maximise contrast for the reader. According to Cairo (2013), sequential scales are most effective in representing quantitative density, as the human eye visually assesses darker hues as being "more". This aligns with the Martini Glass model (Segel and Heer, 2010); the viewer can initially assess the overall darker hues at a glance, which are highlighted through this gradient method, and then utilise the interactive scrubber to look at individual track performance metrics.

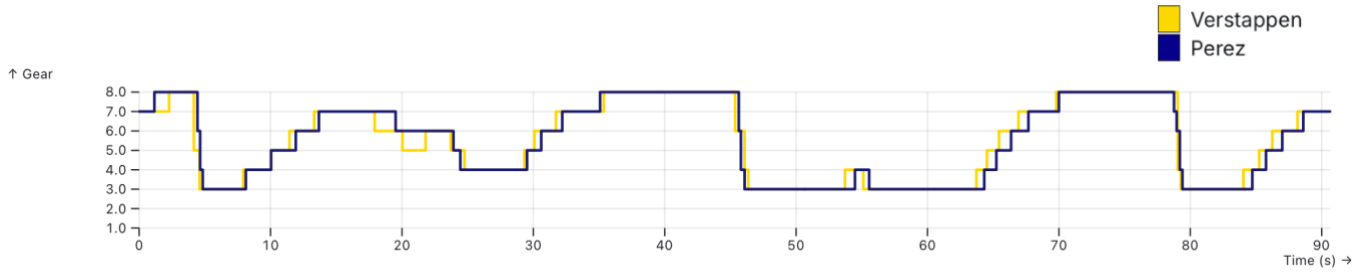
Rejected Attempt 1: Overlapping Data

The strip plot was rejected because both braking zones were on the same temporal plane, and thus the gold and dark blue ticks masked the individual performance profiles for each driver, creating an increased cognitive load (Ware, 2004), resulting in a need to use spatial juxtaposition to provide a discrete, measurable comparison.



Rejected Attempt 2: Cognitive Load

The high information density of the gear step plot resulted in the overwhelming cognitive load that led to its rejection (Desbarats, 2023). Although accurate in a technical sense, the crossed lines of the gear step plot created a complex visual maze that obscured the main narrative of 107% performance. The decision was made to focus on cleaner speed, throttle and brake telemetry for better causal clarity.



5. Validation

The telemetry layout's primary advantage was its adherence to the perceptual hierarchy (Cleveland and McGill, 1985), using a common scale to position all driver inputs, which allows for a more accurate comparison. In contrast to the rejected overlapping strip chart, in which brake application was difficult and unclear to interpret, the final telemetry layout provided a clear representation for all metrics. Although the presentation requires greater cognitive effort to decode than the simplified barcode chart, it provides sufficient granularity to assist in determining performance gaps. In effect, the 107% threshold serves as an anchor (Cairo, 2013) to provide an objective reference point to convert abstract telemetry data into an indicator of regulatory violation.

The logical sequence of the visualisations is a core strength because it leads to a natural progression from the macro view (season trends associated with the heatmap and strip plot) to the micro view (details associated with the mechanics of one lap). The macro-to-micro flow gives clarity to the 107% narrative and keeps it focused and easy to understand. The downside of this approach is its lack of adaptability due to its heavy reliance on car telemetry and ignoring all external variables (e.g., track surface, grip, and dirty air due to other cars' aerodynamics). While the dashboard provides a detailed description of what occurred based on mechanical factors and performance on track layouts, the lack of any external or situational data makes it difficult to understand why the car did not finish above the 107% threshold at specific circuits.

Using the Martini Glass model (Segel & Heer, 2010), the dashboard offers a structured entry point through the heatmap and then enables user-led discovery. The high-density initial view provided by this method helps to support the already established 'overview first, zoom and filter, then details on demand' mantra (Shneiderman, 1996). Therefore, this will allow users to do an immediate visual discovery in the “overview” rather than having to first filter their data to find any usable patterns. However, this overall structure creates a noticeable tradeoff between the depth of interaction; currently, “details-on-demand” can only be accessed along a single temporal axis using the scrubber. Although the design provides an excellent narrated presentation, there is currently no ability to “drill down” into secondary drivers of performance (e.g., tyre degradation curves, reliability history), which is necessary for a completely exploratory tool.

6. References

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